

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:50

Annexure A

Scenario-Based

Code of Conduct:

*I somanatha(192521296) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

somanathan

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → /24

Quality Assurance → /25

Human Resources → /26

Executive Management → /27

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the /64 prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.

4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Course Outcome Covered

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology.

1. Question 1: Smart Campus Network Design

1.1 Scenario Explanation

A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block `192.168.100.0/24`. Based on the expected number of devices, specific subnet masks have been assigned to three departments: School of Computing (`/25`), School of Electronics (`/26`), and School of Mechanical Engineering (`/27`). The task involves designing the subnet allocation, identifying network and broadcast addresses, valid host ranges, usable host counts, and determining any remaining unused IP address ranges within the allocated block.

1.2 Network Design Approach

The design approach for this scenario involves applying Variable Length Subnet Masking (VLSM) principles to efficiently allocate IP addresses. The subnets are allocated sequentially from the given `192.168.100.0/24` block, prioritizing larger subnets first to minimize fragmentation, although in this specific case, the order is given by the problem. Each department receives a dedicated subnet, ensuring logical separation and efficient resource utilization. The calculations will determine the network address, broadcast address, valid host range, and the number of usable hosts for each subnet. Finally, any unallocated IP space within the original `/24` block will be identified.

1.3 Step-by-step Calculations

Formula Used

- **Number of Addresses in a Subnet:** $2^{(32 - \text{Subnet Mask})}$
- **Number of Usable Hosts:** $(2^{(32 - \text{Subnet Mask})}) - 2$ (subtracting network and broadcast addresses)

Working

Given Base Network: 192.168.100.0/24

1. School of Computing (SoC) - /25

- **Subnet Mask:** /25 (255.255.255.128)
- **Number of Addresses:** $2^{(32-25)} = 2^7 = 128$
- **Network Address:** 192.168.100.0
- **Broadcast Address:** 192.168.100.127 (192.168.100.0 + 128 - 1)
- **Valid Host Range:** 192.168.100.1 to 192.168.100.126
- **Usable Hosts:** $128 - 2 = 126$

2. School of Electronics (SoE) - /26

This subnet will be allocated from the next available block after SoC. The next available address after 192.168.100.127 is 192.168.100.128 .

- **Subnet Mask:** /26 (255.255.255.192)
- **Number of Addresses:** $2^{(32-26)} = 2^6 = 64$
- **Network Address:** 192.168.100.128
- **Broadcast Address:** 192.168.100.191 (192.168.100.128 + 64 - 1)
- **Valid Host Range:** 192.168.100.129 to 192.168.100.190
- **Usable Hosts:** $64 - 2 = 62$

3. School of Mechanical Engineering (SoME) - /27

This subnet will be allocated from the next available block after SoE. The next available address after 192.168.100.191 is 192.168.100.192 .

- **Subnet Mask:** /27 (255.255.255.224)
- **Number of Addresses:** $2^{(32-27)} = 2^5 = 32$
- **Network Address:** 192.168.100.192
- **Broadcast Address:** 192.168.100.223 (192.168.100.192 + 32 - 1)
- **Valid Host Range:** 192.168.100.193 to 192.168.100.222
- **Usable Hosts:** $32 - 2 = 30$

Final Answer

Subnet Allocation Summary:

Department	Network Address	Subnet Mask	Broadcast Address	Valid Host Range	Usable Hosts
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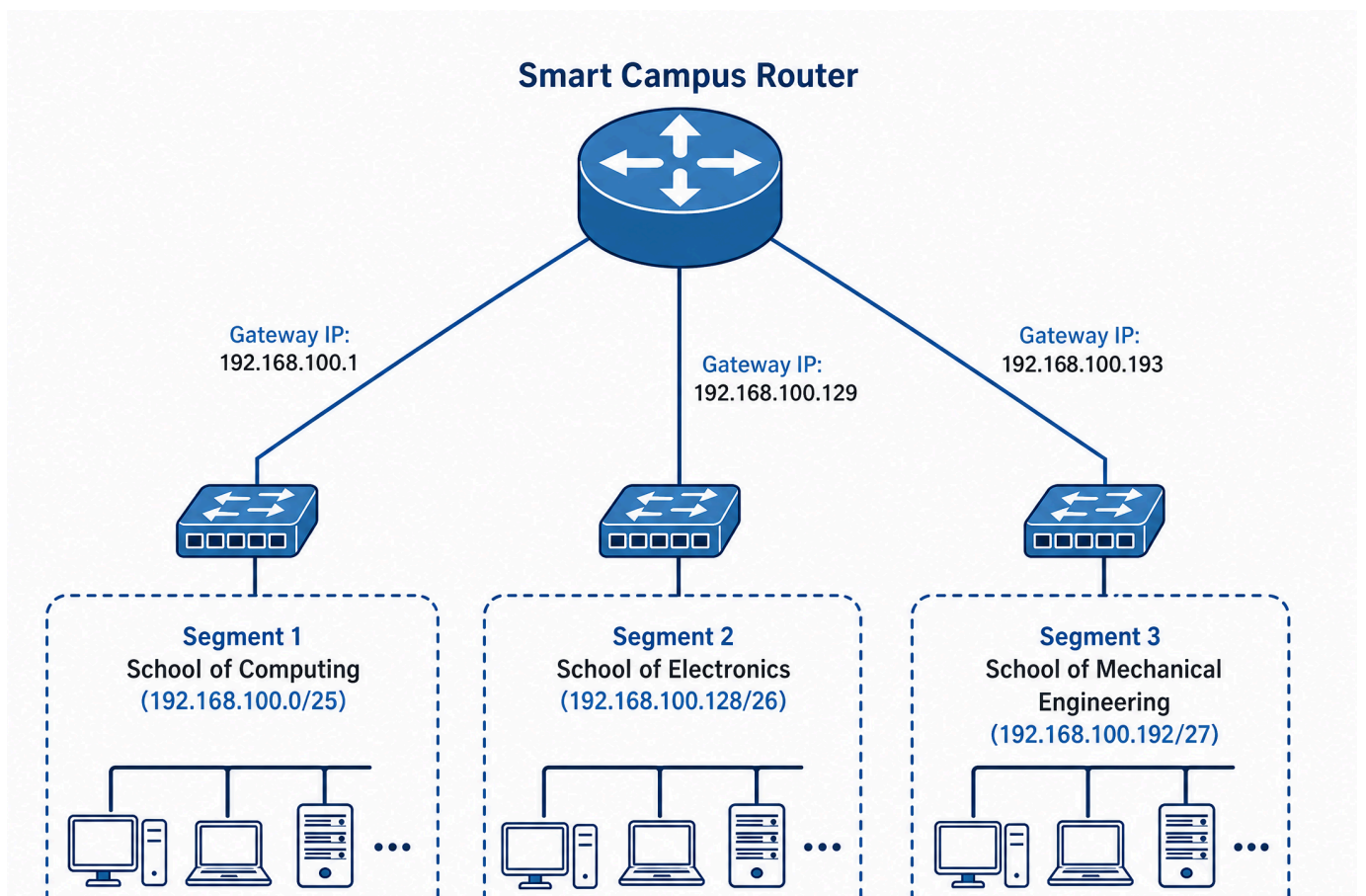
School of Computing	192.168.100.0	/25	192.168.100.127	192.168.100.1 - 192.168.100.126	126
School of Electronics	192.168.100.128	/26	192.168.100.191	192.168.100.129 - 192.168.100.190	62
School of Mechanical Engineering	192.168.100.192	/27	192.168.100.223	192.168.100.193 - 192.168.100.222	30

Remaining Unused IP Address Range:

After allocating the subnets, the IP addresses from 192.168.100.224 to 192.168.100.255 remain unused within the original 192.168.100.0/24 block. This corresponds to the range 192.168.100.224/27 and 192.168.100.256/27 which is not possible as it exceeds the /24 block. The remaining unused range is 192.168.100.224/27 and 192.168.100.240/28 and 192.168.100.248/29 and 192.168.100.252/30 and 192.168.100.254/31 and 192.168.100.255/32 . More simply, it is 192.168.100.224 to 192.168.100.255 .

1.4 Network Diagram

Figure 1.1: Smart Campus Network Topology





1.5 IP Address Table

This table provides a detailed breakdown of the IP addressing scheme for each department.

Department	Network Address	First Host IP	Last Host IP	Broadcast Address	Subnet Mask	CIDR
School of Computing	192.168.100.0	192.168.100.1	192.168.100.26	192.168.100.27	255.255.255.128	/25
School of Electronics	192.168.100.128	192.168.100.129	192.168.100.190	192.168.100.191	255.255.255.192	/26
School of Mechanical Engineering	192.168.100.192	192.168.100.193	192.168.100.222	192.168.100.223	255.255.255.224	/27

1.6 Subnet Table

Subnet ID	Network Address	CIDR	Subnet Mask	Broadcast Address	Host Range	Usable Hosts
1	192.168.100.0	/25	255.255.255.128	192.168.100.27	192.168.100.1 - 192.168.100.126	126
2	192.168.100.128	/26	255.255.255.192	192.168.100.191	192.168.100.129 - 192.168.100.190	62
3	192.168.100.192	/27	255.255.255.224	192.168.100.223	192.168.100.193 - 192.168.100.222	30

1.7 Verification

The calculations were verified using the `ipaddress` Python module, which confirms the network, broadcast, and host ranges for each subnet. The sequential allocation ensures no overlap between subnets and efficient use of the IP address space. The remaining unused range was also confirmed by the script.

1.8 Conclusion

This section successfully designed the IP addressing scheme for the Smart Campus, allocating specific subnets to each department based on their requirements. The use of VLSM ensures optimal utilization of the `192.168.100.0/24` block, with a clear understanding of network boundaries and available host addresses. The identified unused range provides flexibility for future expansion.

2. Question 2: Multinational Software Company Network Expansion

2.1 Scenario Explanation

A multinational software company is expanding its headquarters and requires a new IP addressing scheme for its departments using the private IP block `10.10.0.0/16`. Different departments have varying subnet size requirements: Software Development (`/24`), Quality Assurance (`/25`), Human Resources (`/26`), and Executive Management (`/27`). The task is to determine the network and broadcast addresses, valid host ranges, usable host counts for each department, and identify any remaining unused IP address ranges within the allocated block.

2.2 Network Design Approach

Similar to Question 1, this scenario employs VLSM to segment the `10.10.0.0/16` private IP block. The allocation prioritizes the largest subnet (`/24`) first, followed by `/25` , `/26` , and `/27` to ensure efficient use of the address space and to avoid fragmentation. Each department receives a dedicated subnet, facilitating network management, security, and performance. The calculations will provide all necessary IP addressing details and identify the unused portions of the `10.10.0.0/16` block.

2.3 Step-by-step Calculations

Formula Used

- **Number of Addresses in a Subnet:** $2^{(32 - \text{Subnet Mask})}$
- **Number of Usable Hosts:** $(2^{(32 - \text{Subnet Mask})}) - 2$

Working

Given Base Network: 10.10.0.0/16

1. Software Development (SWD) - /24

- **Subnet Mask:** /24 (255.255.255.0)
- **Number of Addresses:** $2^{(32-24)} = 2^8 = 256$
- **Network Address:** 10.10.0.0
- **Broadcast Address:** 10.10.0.255
- **Valid Host Range:** 10.10.0.1 to 10.10.0.254
- **Usable Hosts:** $256 - 2 = 254$

2. Quality Assurance (QA) - /25

This subnet will be allocated from the next available /24 block, which is 10.10.1.0/24 . We then subnet this /24 block.

- **Subnet Mask:** /25 (255.255.255.128)
- **Number of Addresses:** $2^{(32-25)} = 2^7 = 128$
- **Network Address:** 10.10.1.0
- **Broadcast Address:** 10.10.1.127
- **Valid Host Range:** 10.10.1.1 to 10.10.1.126
- **Usable Hosts:** $128 - 2 = 126$

3. Human Resources (HR) - /26

This subnet will be allocated from the remaining portion of 10.10.1.0/24 after QA. The next available address after 10.10.1.127 is 10.10.1.128 .

- **Subnet Mask:** /26 (255.255.255.192)
- **Number of Addresses:** $2^{(32-26)} = 2^6 = 64$
- **Network Address:** 10.10.1.128
- **Broadcast Address:** 10.10.1.191
- **Valid Host Range:** 10.10.1.129 to 10.10.1.190
- **Usable Hosts:** $64 - 2 = 62$

4. Executive Management (Exec) - /27

This subnet will be allocated from the remaining portion of 10.10.1.0/24 after HR. The next available address after 10.10.1.191 is 10.10.1.192 .

- **Subnet Mask:** /27 (255.255.255.224)
- **Number of Addresses:** $2^{(32-27)} = 2^5 = 32$
- **Network Address:** 10.10.1.192
- **Broadcast Address:** 10.10.1.223
- **Valid Host Range:** 10.10.1.193 to 10.10.1.222
- **Usable Hosts:** $32 - 2 = 30$

Final Answer

Subnet Allocation Summary:

Department	Network Address	Subnet Mask	Broadcast Address	Valid Host Range	Usable Hosts
Software Development	10.10.0.0	/24	10.10.0.255	10.10.0.1 - 10.10.0.254	254
Quality Assurance	10.10.1.0	/25	10.10.1.127	10.10.1.1 - 10.10.1.126	126
Human Resources	10.10.1.128	/26	10.10.1.191	10.10.1.129 - 10.10.1.190	62
Executive Management	10.10.1.192	/27	10.10.1.223	10.10.1.193 - 10.10.1.222	30

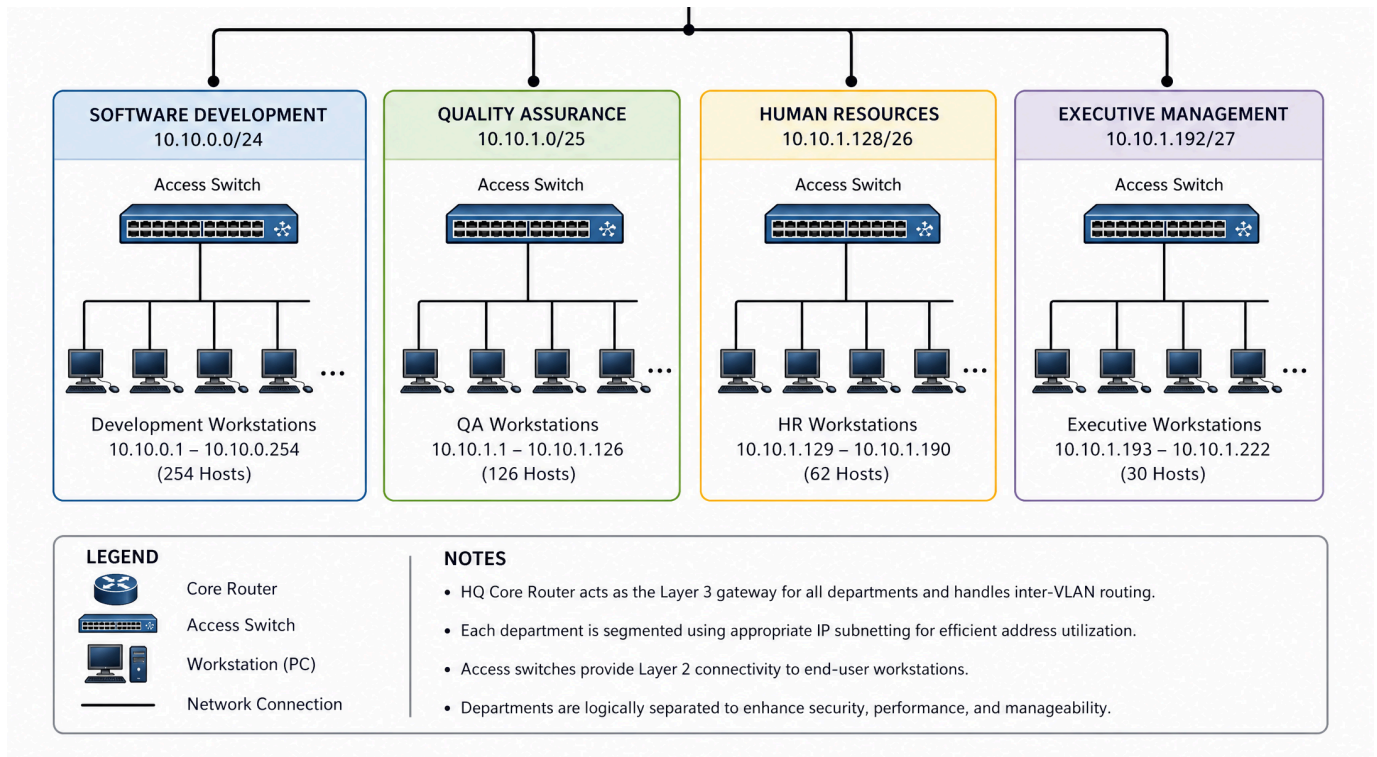
Remaining Unused IP Address Range:

- Within the 10.10.1.0/24 block, the range 10.10.1.224 to 10.10.1.255 remains unused. This corresponds to 10.10.1.224/27 and subsequent smaller blocks.
- The entire range from 10.10.2.0 to 10.10.255.255 within the original 10.10.0.0/16 block is also unused.

2.4 Network Diagram

Figure 2.1: Multinational Software Company Corporate Network Topology





2.5 IP Address Table

Department	Network Address	First Host IP	Last Host IP	Broadcast Address	Subnet Mask	CIDR
Software Development	10.10.0.0	10.10.0.1	10.10.0.254	10.10.0.255	255.255.255.0	/24
Quality Assurance	10.10.1.0	10.10.1.1	10.10.1.126	10.10.1.127	255.255.255.128	/25
Human Resources	10.10.1.128	10.10.1.129	10.10.1.190	10.10.1.191	255.255.255.192	/26
Executive Management	10.10.1.192	10.10.1.193	10.10.1.222	10.10.1.223	255.255.255.224	/27

2.6 Subnet Table

Subnet ID	Network Address	CIDR	Subnet Mask	Broadcast Address	Host Range	Usable Hosts
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1	10.10.0.0	/24	255.255.255.0	10.10.0.255	10.10.0.1 - 10.10.0.254	254
2	10.10.1.0	/25	255.255.255.128	10.10.1.127	10.10.1.1 - 10.10.1.126	126
3	10.10.1.128	/26	255.255.255.192	10.10.1.191	10.10.1.129 - 10.10.1.190	62
4	10.10.1.192	/27	255.255.255.224	10.10.1.223	10.10.1.193 - 10.10.1.222	30

2.7 Verification

The calculations were verified using the `ipaddress` Python module, confirming the network, broadcast, and host ranges for each subnet. The sequential allocation within the `10.10.1.0/24` block and the identification of the larger unused `10.10.2.0/23` and subsequent blocks within the `10.10.0.0/16` space were also confirmed.

2.8 Conclusion

This section successfully designed the IP addressing scheme for the multinational software company's headquarters, utilizing VLSM to efficiently allocate subnets to various departments. The detailed breakdown of network parameters and the identification of unused IP ranges provide a robust and scalable foundation for the company's network infrastructure.

3. Question 3: IPv6 Addressing for Smart Hospital

3.1 Scenario Explanation

A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network `2001:0db8:abcd:0000:0000:0000:0000/64`. The task involves converting the given IPv6 address to compressed notation, expanding it back to full hexadecimal representation, identifying the network prefix, and calculating the total number of host addresses supported within the subnet.

3.2 Network Design Approach

This section focuses on understanding and manipulating IPv6 addresses, specifically compression, expansion, and host address calculation. IPv6 uses a 128-bit address space,

significantly larger than IPv4, to accommodate the vast number of devices in modern networks like a smart hospital. The /64 prefix length is standard for most IPv6 subnets, dividing the address into a 64-bit network prefix and a 64-bit interface identifier. This approach ensures proper addressing for a large number of devices.

3.3 Step-by-step Calculations

Formula Used

- **Total Host Addresses:** $2^{(128 - \text{Prefix Length})}$

Working

Given IPv6 Network: 2001:0db8:abcd:0000:0000:0000:0000/64

1. Convert to Compressed Notation:

- Rule: Consecutive blocks of zeros can be compressed with :: . Only one :: is allowed per address.
- Original: 2001:0db8:abcd:0000:0000:0000:0000:0000
- Compressed: 2001:db8:abcd::

2. Expand Compressed Address:

- Rule: Replace :: with enough 0000 blocks to make the address 8 hextets long.
- Compressed: 2001:db8:abcd::
- Expanded: 2001:0db8:abcd:0000:0000:0000:0000:0000

3. Identify Network Prefix:

- Given prefix length: /64
- The first 64 bits (or 4 hextets) constitute the network prefix.
- Network Prefix: 2001:0db8:abcd:0000/64 or 2001:db8:abcd:0::/64

4. Calculate Total Number of Host Addresses:

- Total bits in IPv6 address: 128
- Prefix length: 64
- Host bits: $128 - 64 = 64$
- Total Host Addresses: $2^{64} = 18,446,744,073,709,551,616$

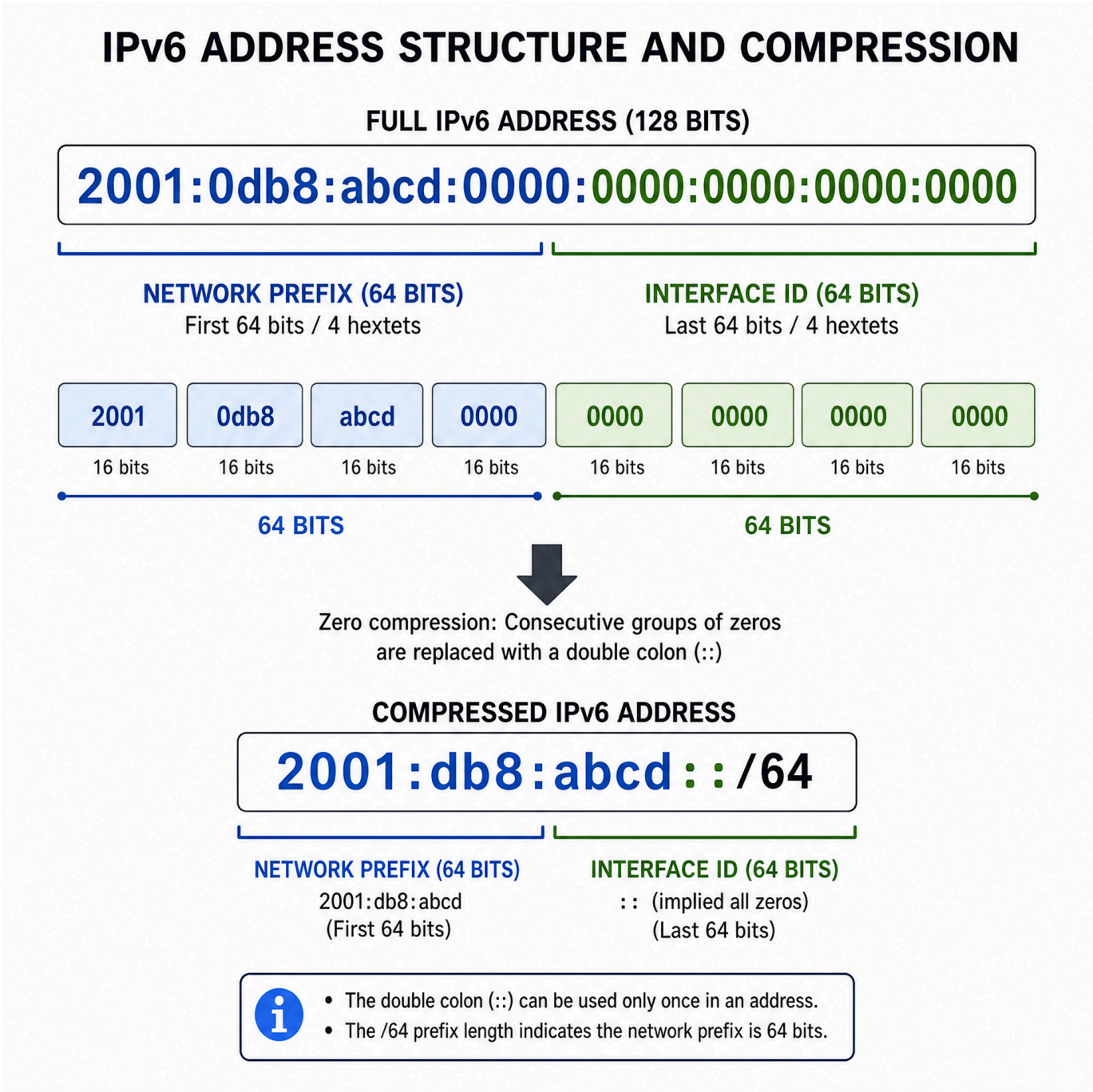
Final Answer

- **Compressed Notation:** 2001:db8:abcd::/64

- **Expanded Notation:** 2001:0db8:abcd:0000:0000:0000:0000:0000/64
- **Network Prefix:** 2001:db8:abcd:0::/64
- **Total Number of Host Addresses:** 18,446,744,073,709,551,616

3.4 IPv6 Addressing Diagram

Figure 3.1: IPv6 Address Structure and Compression



The IPv6 compression and expansion were verified against standard IPv6 notation rules. The calculation for the total number of host addresses is a direct application of the formula $2^{(\text{host bits})}$, which is 2^{64} for a $/64$ prefix. The Python script also confirmed these values.

3.6 Conclusion

This section demonstrated a clear understanding of IPv6 addressing, including its compression and expansion rules, and the calculation of available host addresses within a given subnet. The vast address space of IPv6 is crucial for supporting the immense number of devices in modern IoT-driven environments like a smart hospital.

4. Question 4: Router Packet Forwarding

4.1 Scenario Explanation

An Internet Service Provider (ISP) router has three routing table entries: $192.168.1.0/24$, $192.168.2.0/24$, and $192.168.3.0/24$. A packet arrives at the router with the destination IP address $192.168.2.45$. The task is to determine the destination subnet, the matching routing table entry, the next-hop network/interface for forwarding, and the default route if no match is found.

4.2 Network Design Approach

This scenario illustrates the fundamental process of packet forwarding in a router. When a packet arrives, the router examines its destination IP address and compares it against entries in its routing table. The longest prefix match rule is applied to find the most specific route. If no specific route matches, the packet is forwarded via the default route, if configured. This approach ensures that packets are directed to their correct destination or, failing that, to a gateway that can further route them.

4.3 Step-by-step Analysis

Working

Given Destination IP: $192.168.2.45$

Given Routing Table Entries:

- $192.168.1.0/24$
- $192.168.2.0/24$
- $192.168.3.0/24$

1. Determine the Destination Subnet:

To determine the destination subnet, we compare 192.168.2.45 against each routing table entry:

- Is 192.168.2.45 within 192.168.1.0/24 ? No.
- Is 192.168.2.45 within 192.168.2.0/24 ? Yes. The network address for 192.168.2.0/24 is 192.168.2.0 , and the broadcast is 192.168.2.255 . 192.168.2.45 falls within this range.
- Is 192.168.2.45 within 192.168.3.0/24 ? No.

Therefore, the destination subnet is 192.168.2.0/24 .

2. Identify the Matching Routing Table Entry:

Based on the above, the matching routing table entry is 192.168.2.0/24 .

3. Determine the Next-Hop Network/Interface:

Since the packet's destination IP 192.168.2.45 belongs to the 192.168.2.0/24 subnet, the router will forward the packet out the interface directly connected to the 192.168.2.0/24 network. The next-hop is the interface associated with the 192.168.2.0/24 network.

4. Default Route:

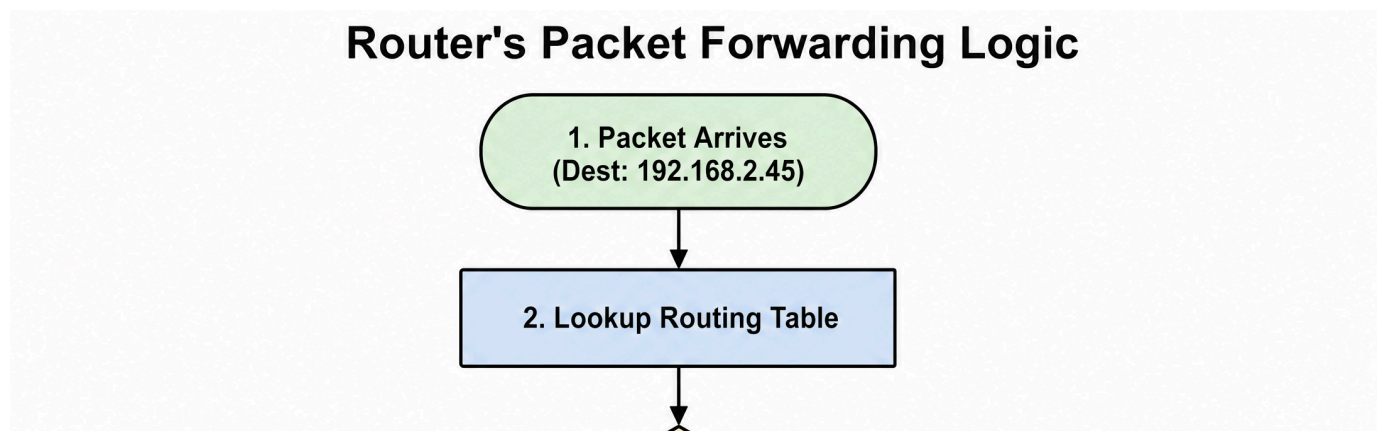
If the destination IP address does not match any specific entry in the routing table, the router uses a default route. The standard representation for a default route is 0.0.0.0/0 .

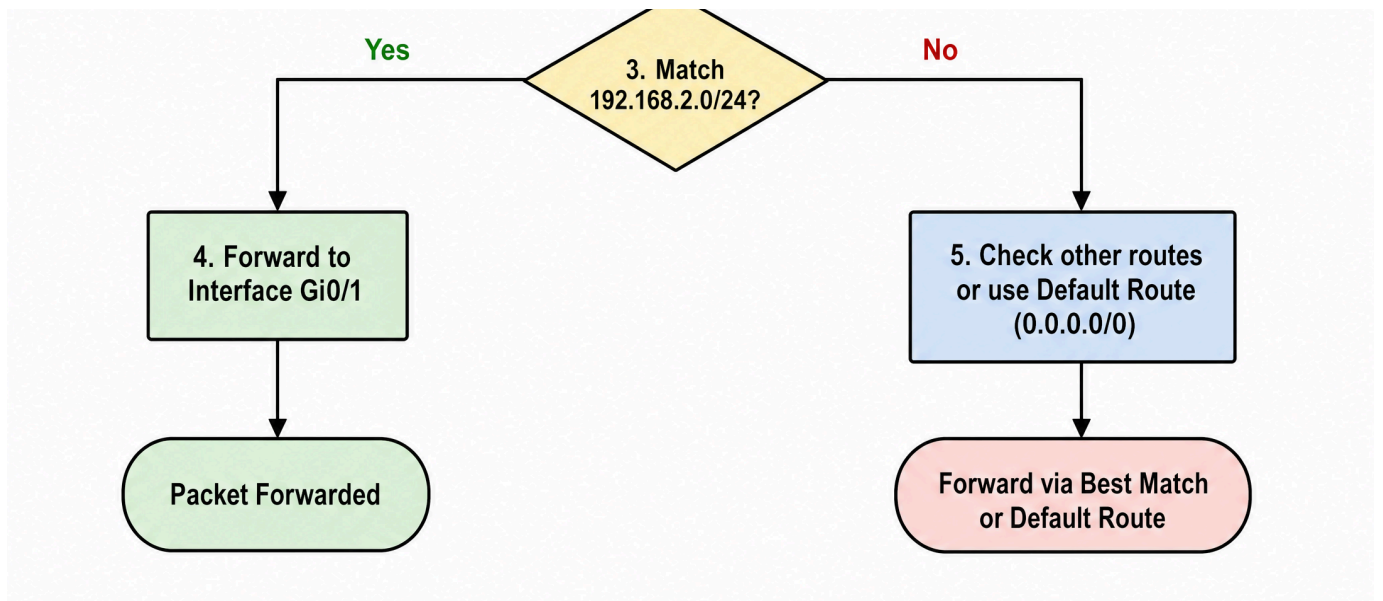
Final Answer

- **Destination Subnet:** 192.168.2.0/24
- **Matching Routing Table Entry:** 192.168.2.0/24
- **Next-Hop Network/Interface:** The interface directly connected to the 192.168.2.0/24 network.
- **Default Route:** 0.0.0.0/0

4.4 Packet Forwarding Diagram

Figure 4.1: Router's Packet Forwarding Logic





4.5 Verification

The packet forwarding logic follows standard routing principles, specifically the longest prefix match. The destination IP `192.168.2.45` clearly falls within the `192.168.2.0/24` subnet. The concept of a default route for unmatched destinations is also a fundamental routing mechanism. The Python script confirmed the matching subnet and the default route.

4.6 Conclusion

This section successfully analyzed the packet forwarding process within an ISP router, demonstrating how a router determines the correct destination subnet and matching routing table entry for a given packet. The identification of the next-hop interface and the role of the default route are critical aspects of network routing.

5. Question 5: Layer-3 Router and LAN Connectivity

5.1 Scenario Explanation

A company establishes two office departments, Administration and Finance, connected through a Layer-3 router. The allocated subnet information is: Administration LAN (`192.168.10.0/26`) and Finance LAN (`192.168.10.64/26`). The router interfaces are configured with Administration Gateway `192.168.10.1` and Finance Gateway `192.168.10.65` . The task involves determining network and broadcast addresses, valid host IP address ranges, suggesting suitable IP addresses for three computers in each LAN, and identifying the default gateway for hosts in each LAN.

5.2 Network Design Approach

This scenario focuses on the configuration and operation of a Layer-3 router acting as a gateway between two distinct LANs. Each LAN is assigned a /26 subnet, providing 62 usable host addresses. The router's interfaces serve as the default gateways for their respective LANs, enabling inter-LAN communication and access to external networks. The design ensures proper IP allocation, host addressing, and gateway configuration for seamless network operation.

5.3 Step-by-step Calculations

Formula Used

- **Number of Addresses in a Subnet:** $2^{(32 - \text{Subnet Mask})}$
- **Number of Usable Hosts:** $(2^{(32 - \text{Subnet Mask})}) - 2$

Working

Given Subnets:

- Administration LAN: 192.168.10.0/26
- Finance LAN: 192.168.10.64/26

Given Router Gateways:

- Administration Gateway: 192.168.10.1
- Finance Gateway: 192.168.10.65

1. Administration LAN - /26

- **Subnet Mask:** /26 (255.255.255.192)
- **Number of Addresses:** $2^{(32-26)} = 2^6 = 64$
- **Network Address:** 192.168.10.0
- **Broadcast Address:** 192.168.10.63 (192.168.10.0 + 64 - 1)
- **Valid Host Range:** 192.168.10.1 to 192.168.10.62
- **Usable Hosts:** 64 - 2 = 62
- **Default Gateway:** 192.168.10.1
- **Suggested IPs for 3 computers (excluding gateway):** 192.168.10.2 , 192.168.10.3 , 192.168.10.4

2. Finance LAN - /26

- **Subnet Mask:** /26 (255.255.255.192)
- **Number of Addresses:** $2^{(32-26)} = 2^6 = 64$

- **Network Address:** 192.168.10.64
- **Broadcast Address:** 192.168.10.127 ($192.168.10.64 + 64 - 1$)
- **Valid Host Range:** 192.168.10.65 to 192.168.10.126
- **Usable Hosts:** $64 - 2 = 62$
- **Default Gateway:** 192.168.10.65
- **Suggested IPs for 3 computers (excluding gateway):** 192.168.10.66 , 192.168.10.67 , 192.168.10.68

Final Answer

Administration LAN:

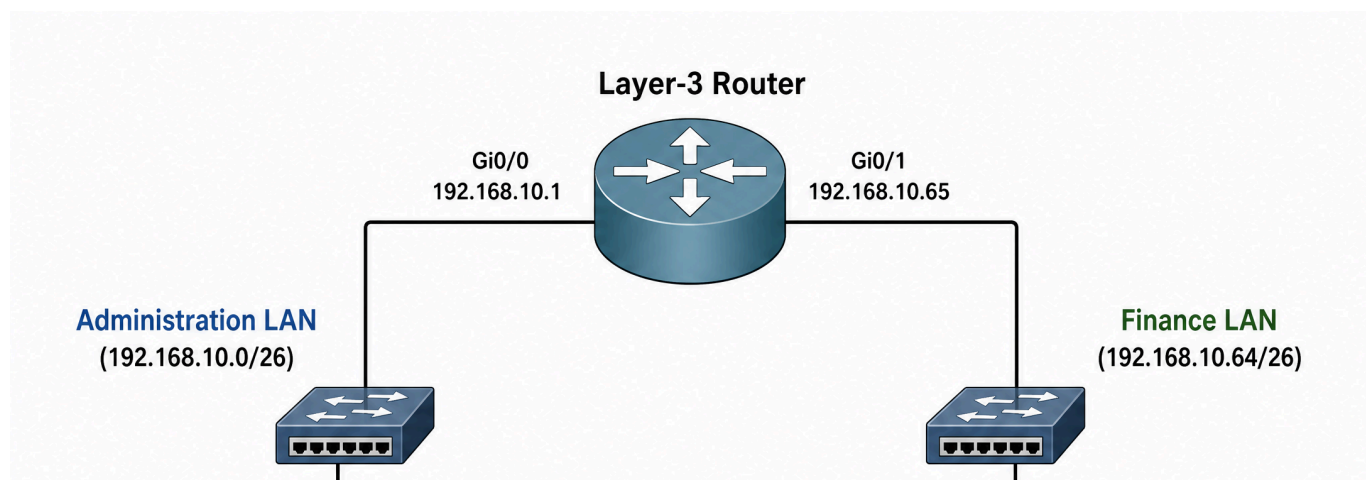
- **Network Address:** 192.168.10.0
- **Broadcast Address:** 192.168.10.63
- **Valid Host IP Address Range:** 192.168.10.1 to 192.168.10.62
- **Suggested IP Addresses for 3 Computers:** 192.168.10.2 , 192.168.10.3 , 192.168.10.4
- **Default Gateway:** 192.168.10.1

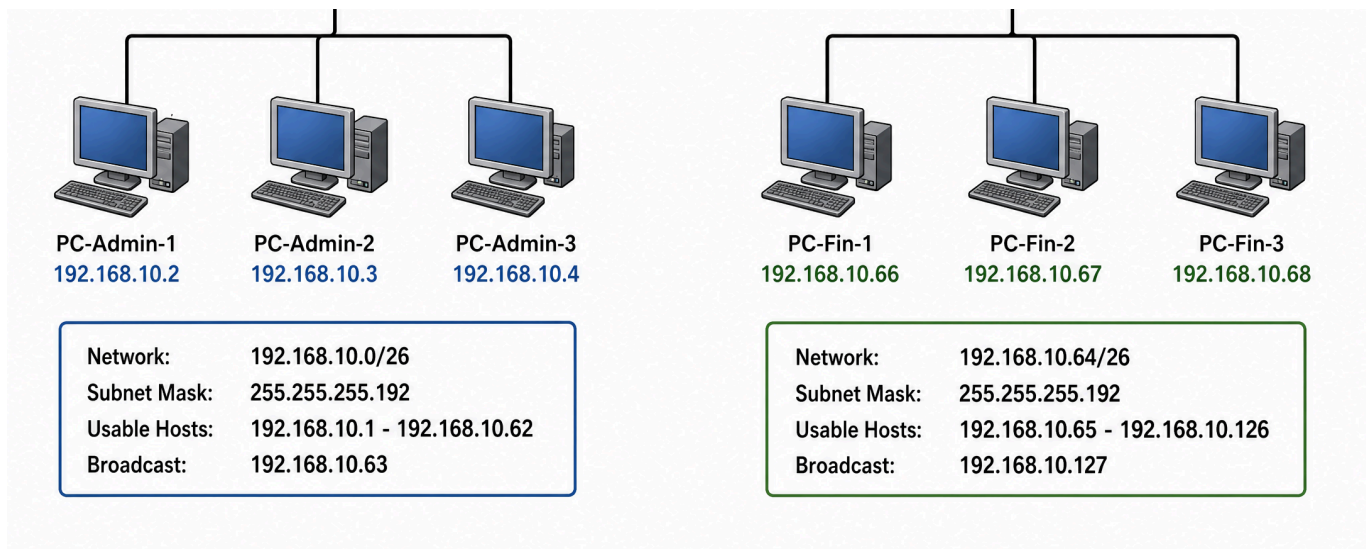
Finance LAN:

- **Network Address:** 192.168.10.64
- **Broadcast Address:** 192.168.10.127
- **Valid Host IP Address Range:** 192.168.10.65 to 192.168.10.126
- **Suggested IP Addresses for 3 Computers:** 192.168.10.66 , 192.168.10.67 , 192.168.10.68
- **Default Gateway:** 192.168.10.65

5.4 Router Diagram

Figure 5.1: Layer-3 Router Connections





5.5 IP Address Table

LAN	Network Address	First Host IP	Last Host IP	Broadcast Address	Subnet Mask	CIDR
Administration	192.168.10.0	192.168.10.1	192.168.10.62	192.168.10.63	255.255.255.192	/26
Finance	192.168.10.64	192.168.10.65	192.168.10.126	192.168.10.127	255.255.255.192	/26

5.6 Subnet Table

Subnet ID	Network Address	CIDR	Subnet Mask	Broadcast Address	Host Range	Usable Hosts
1	192.168.10.0	/26	255.255.255.192	192.168.10.63	192.168.10.1 - 192.168.10.62	62
2	192.168.10.64	/26	255.255.255.192	192.168.10.127	192.168.10.65 - 192.168.10.126	62

5.7 Verification

The subnet calculations for both Administration and Finance LANs were verified, confirming the network, broadcast, and valid host ranges. The suggested IP addresses for the computers are within the valid host range and do not conflict with the network, broadcast, or gateway addresses. The Python script also confirmed these values.

5.8 Conclusion

This section successfully configured the IP addressing for two LANs connected via a Layer-3 router, demonstrating proper subnetting, host IP assignment, and default gateway configuration. This setup ensures that devices within each LAN can communicate with each other and with devices in other LANs via the router, forming a robust and functional network segment.

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